FoodWise AI

AI-Powered Food Waste Prediction,
Surplus Detection & Redistribution Support Platform

Concept Note • IBM SkillsBuild Internship • SDG 12 — Responsible Consumption & Production

1. Executive Summary

FoodWise AI is an AI-powered **decision-support prototype** designed to help food-serving organisations — including canteens, restaurants, hotel food-service departments, and institutional kitchens — predict food demand, detect potential surplus risk early, and recommend appropriate actions before food becomes waste.

The application is built using Python, Streamlit, scikit-learn, and Plotly. It demonstrates a complete data-driven pipeline: from CSV-based data input and automated cleaning, through machine-learning demand forecasting, transparent risk scoring, five-category action recommendations, and simulated redistribution matching. A sustainability insights dashboard presents all key metrics in the context of **SDG 12 — Responsible Consumption and Production, Target 12.3**.

The prototype passed a 100-check functional QA audit with **98 PASS, 0 FAIL**. The Random Forest demand prediction model achieved $R^2 = 0.797$, MAE = 5.80 units, and RMSE = 7.49 units on the project's synthetic/demo dataset. These are *prototype evaluation results on synthetic data* and do not represent production performance.

2. Problem Statement

Food waste is a major global sustainability challenge. Approximately one-third of all food produced for human consumption is lost or wasted each year (UN FAO). Food-serving organisations frequently over-prepare meals due to demand uncertainty, generating avoidable surplus that often ends up discarded.

The key contributing factors include:

- Demand uncertainty — staff rely on manual estimation or intuition, with no systematic prediction tool
- Late surplus detection — remaining food is identified only after service, when options are limited
- No structured decision framework — staff lack a data-driven workflow for surplus decisions (discount, redistribute, reduce preparation)
- Missed redistribution opportunities — surplus that could be shared is discarded due to time constraints or lack of matching infrastructure

3. Background and Context

The United Nations Sustainable Development Goal 12 — Responsible Consumption and Production — calls for sustainable management and efficient use of natural resources. Its Target 12.3 specifically calls for halving per-capita global food waste at the retail and consumer levels and reducing food losses along production and supply chains by 2030.

At the operational level, food-serving organisations generate daily records of what was prepared and sold. This data, if analysed systematically, can support earlier identification of surplus risk and better decisions about preparation quantities and surplus handling. AI and machine learning techniques offer a practical way to extract actionable insights from this historical data.

4. SDG Alignment

PRIMARY	**SDG 12 — Responsible Consumption and Production**	Target 12.3: Reduce food waste and food losses

POTENTIAL SECONDARY	**SDG 2 — Zero Hunger**	Redistribution support may help eligible surplus reach community organisations

■■ *SDG 2 is described only as a potential secondary contribution. FoodWise AI does not claim to solve hunger, and no actual redistribution is performed by the prototype. Food safety must be independently verified.*

5. Target Users

FoodWise AI is designed for food-serving organisations that prepare food daily and generate preparation records:

- University and college cafeterias
- Restaurants and fast-food outlets
- Hotel food and beverage departments
- Institutional and corporate canteens
- Supermarkets managing perishable inventory
- Catering companies and event kitchens

Early adopters are organisations with high daily food volumes, measurable surplus patterns, and staff responsible for both preparation planning and surplus disposal decisions.

6. Proposed Solution

FoodWise AI addresses the identified problem through an integrated data-driven pipeline:

Step	Component	Description
1	**Data Input**	Load bundled sample CSV or upload real food-preparation records
2	**Cleaning & Validation**	Detect and fix date errors, negatives, impossible quantities; report quality metrics
3	**Analytics**	Interactive charts: demand trends, category breakdown, day-of-week patterns, top-surplus items
4	**Demand Prediction**	Random Forest Regressor predicts Sold_Quantity per item; outputs Predicted_Demand and Predicted_
5	**Waste Risk Scoring**	Transparent Low/Medium/High risk using Waste_Ratio and Expiry_Hours; escalation for short expiry
6	**Action Recommendations**	Five-label engine: Monitor, Reduce Preparation, Controlled Discount, Redistribution Candidate, Discard
7	**Redistribution Support**	Matches redistribution-candidate records to simulated recipient organisations by category and capacity
8	**Sustainability Insights**	SDG-aligned KPI dashboard: estimated surplus, risk counts, redistribution candidates, model metrics

7. AI / Machine Learning Approach

The demand prediction module uses a **Random Forest Regressor** (scikit-learn RandomForestRegressor: 100 trees, max depth 10, random state 42). The model predicts *Sold_Quantity* for a given food item on a given date.

Input Features

- Food_Item (label-encoded) — which item is being prepared
- Category (label-encoded) — food category (Grains, Protein, Vegetables, etc.)
- DayOfWeek — 0 = Monday, 6 = Sunday
- Month — 1–12
- IsWeekend — 1 if Saturday or Sunday, else 0

Evaluation Approach

A **time-based 80/20 train/test split** is used: earlier dates form the training set, later dates form the test set. This avoids data leakage from future observations into the model.

Prototype Evaluation Results

Mean Absolute Error (MAE)	**5.80 units**	Average absolute difference between predicted and actual sold quantity
Root Mean Squared Error (RMSE)	**7.49 units**	Penalises larger errors more heavily than MAE
R^2 (Coefficient of Determination)	**0.797**	~80% of variance in sold quantity explained by the five features

■■ *These results were obtained entirely on the project's synthetic/demo dataset and must not be interpreted as production forecasting performance or real-world accuracy. Performance on real operational data will differ.*

8. Core Features

Data Input & Validation	Loads bundled sample dataset (6,576 synthetic records) or user-uploaded CSV. Validates required columns; reports missing columns clearly.
Data Cleaning	Handles unparseable dates, non-numeric quantities, negatives (clamped to 0), Sold > Prepared (capped), and Remaining always recalculated as Prepared – Sold.
Analytics Dashboard	Interactive Plotly charts: daily trends, category breakdown, day-of-week patterns, top-surplus items, monthly surplus trend. Date-range filter included.
Demand Prediction	Random Forest Regressor trained on food item, category, and date features. Outputs Predicted_Demand and Predicted_Surplus per record.
Surplus & Waste Risk	Transparent Low/Medium/High risk scoring using Waste_Ratio = Remaining ÷ Prepared and Expiry_Hours. Risk is independent of Food_Safety_Status.
Action Recommendations	Five-label engine per record: Monitor, Reduce Preparation, Controlled Discount, Redistribution Candidate (Safe+High only), or Discard (Unsafe only).
Redistribution Support	Matches redistribution-candidate records to simulated recipient organisations (10 fictional orgs) by food category and daily capacity. Prototype only.
Sustainability Insights	SDG 12.3-aligned KPI dashboard with estimated surplus totals, risk distribution, redistribution candidates, model metrics, and surplus trend chart.

9. Expected Benefits

As a decision-support tool, FoodWise AI is *designed to support* the following outcomes in a real operational deployment:

- Earlier identification of potential surplus, enabling timely action
- Data-driven preparation planning to reduce over-preparation over time
- Structured decision support for staff managing surplus (discount, prepare less, redistribute)
- A transparent audit trail of surplus events and recommended actions
- A foundation for sustainable food management practices aligned with SDG 12.3

■■ *These are designed capabilities of the prototype. The application does not claim to have already reduced real-world food waste, performed redistribution, or produced verified environmental impact.*

10. Innovation and Value Proposition

FoodWise AI's core innovation is the integration of four capabilities into one lightweight platform:

- ML demand forecasting — moves beyond manual estimation
- Transparent risk scoring — every decision is explainable using visible thresholds
- Structured action recommendations — staff receive a specific suggested action, not just a number
- Redistribution matching — connects eligible surplus with simulated community recipients

The application is built on an entirely open-source Python stack, requires no cloud infrastructure for local deployment, and is designed to be explainable to non-technical decision-makers.

11. Technology Stack

Component	Technology	Version
Web interface	Streamlit	1.65.0
Data manipulation	Pandas	3.0.6
Numerical computing	NumPy	2.5.3
Machine learning	scikit-learn	1.9.1
Visualisation	Plotly	7.1.0
Model persistence	Joblib	1.6.0
Python runtime	CPython	3.12.14
Environment manager	uv	—

12. Data Handling

Sample Dataset

The application includes a **synthetic** sample dataset (*data/sample_food_data.csv*) generated specifically for this prototype. It contains **6,576 records** covering 24 food items across 6 categories (Grains & Bread, Vegetables, Protein, Snacks, Desserts, Beverages) for the period 1 January 2024 – 30 September 2024.

Dataset fields: Date, Food_Item, Category, Prepared_Quantity, Sold_Quantity, Remaining_Quantity, Expiry_Hours, Food_Safety_Status, Price (■). All values are synthetic and do not represent any real organisation or operation.

Redistribution Recipient Data

Ten fictional recipient organisations are included in *data/sample_recipients.csv*. Organisations cover types including NGO / Food Bank, Shelter / Welfare Home, Orphanage, and Community Centre, across six Indian cities. All organisations, contact details, and capacity figures are entirely simulated.

■■ *Redistribution matching is a prototype demonstration. FoodWise AI does not connect to any real NGO, food bank, or community network, and does not perform actual redistribution.*

13. Prototype Results and QA Status

QA Check	Result
Total automated checks	100
PASS	98
FAIL	**0**

Informational warnings	2 (non-blocking, design-confirmed)
Streamlit health check	HTTP 200 — application running
MVP status	Ready for demonstration

14. Limitations

- All results are based on a **synthetic/demo dataset** and cannot be used to make real-world claims
- No real-time integration with restaurant POS, inventory management, or IoT sensors
- No integration with real NGO, food-bank, or redistribution networks
- Food safety is **not** autonomously certified — the Food_Safety_Status field is a rule-based demo label
- ML model trained on synthetic data — real-world performance requires real operational data
- Does not account for food weight, portion sizes, or caloric content
- Redistribution matching is simulated — capacity figures and categories are fictional
- Recommendation engine is a decision-support tool; all suggestions require human review
- No multi-location, multi-organisation, or user authentication support in current prototype

15. Future Scope

- Integration with operational POS, inventory, or ERP systems for real data
- Improved forecasting: LSTM / time-series models with weather and event features
- Real-time inventory monitoring via IoT sensors or API polling
- Verified partnerships with real NGOs and food banks
- Geospatial distance-based recipient matching
- Responsible IBM watsonx / LLM explanation layer for recommendations
- Automated PDF/Excel reporting for management review
- Role-based dashboards for canteen staff vs. managers
- Multi-location and multi-organisation support

16. Conclusion

FoodWise AI demonstrates that a lightweight, open-source Python application can provide meaningful decision support for food waste reduction at the operational level. By combining machine-learning demand forecasting, transparent risk scoring, structured action recommendations, and redistribution matching in a single platform, it offers food-serving organisations a practical starting point for data-driven surplus management.

The prototype is aligned with **SDG 12 — Responsible Consumption and Production, Target 12.3**, and its redistribution support feature represents a potential contribution to SDG 2 in a real-world deployment. It is presented as a *decision-support prototype* with clearly documented capabilities, limitations, and honest evaluation results.

The application passed its full functional QA audit (98/100 checks), is running successfully, and is ready for demonstration.